4.2.1. Month with the Highest Total Sales

05:06

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: `Date`, `Product`, `Quantity`, `Price`, and `City`.
- Group the data by Month and calculate the total sales for each month.
- Find the month with the highest total sales and display it.
- Also, display the total sales for the best month.

Sample Data:

```
Date,Product,Quantity,Price,City
2025-01-01,Product A,5,20,New York
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,8,20,Los Angeles
2025-01-04,Product C,6,30,New York
2025-01-04,Product B,5,15,Los Angeles
2025-01-05,Product A,3,20,Chicago
2025-01-05,Product C,10,30,Los Angeles
```

Note:
The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Code:

```
import pandas as pd



# Prompt the user for the file name

file_name = input()



# Load the data

df = pd.read_csv(file_name)

df["Date"] = pd.to_datetime(df["Date"])

df["Month"] = df["Date"].dt.to_period("M")
```

```
df["Total Sales"] = df["Quantity"] * df["Price"]

monthly_sales=df.groupby("Month")["Total Sales"].sum()

# Find the month with the highest total sales

best_month = monthly_sales.idxmax()

highest_sales = monthly_sales.max()


print(f"Best month: {best_month}")

print(f"Total sales: ${highest_sales:.2f}")
```